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In a pandal, you overturn a table of books only if you can't argue against them. — *Vijay Prashad*
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WORDLY WISE
A word after a word after a word is power.
— *Margaret Atwood*

Fossil fuel in the farm, and food as fuel — a double bind



HARISH DAMODARAN

WAR AND weather are exposing Indian agriculture's vulnerabilities from two fronts — "fossilisation" of farming and "fuelisation" of food.

In 1950-51, the country had hardly 8,500 tractors and its chemical fertiliser consumption was 69,800 tonnes in terms of nitrogen (N), phosphorous (P) and potassium (K). The only fertiliser products were ammonium sulphate (with 20.6 per cent N and 23 per cent sulphur or S content) and single super phosphate (SSP: 16 per cent P and 11 per cent S).

That was a time when farmers overwhelmingly used bullocks for ploughing their fields, treading over harvested crops to separate grain from straw, pulling carts to transport produce, and powering the *rahat* or Persian wheel to lift water from wells for irrigation. Plant nutrients, too, were predominantly supplied by cattle as decomposed farmyard manure with an average 0.5 per cent N, 0.2 per cent P and 0.5 per cent K.

Contrast with the present, where India's tractor population has crossed 1.2 crore and a whopping 11.6 lakh units sold just in 2025-26 (April-March). These have rendered bullocks redundant in field operations and haulage. Harvesting and threshing is increasingly done by combines, while the *rahat* has given way to electric motor and diesel engine-driven pumps.

The same holds for fertiliser consumption, which reached 33.5 million tonnes of N, P and K in 2025-26. Roughly 67.5 per cent of that came from two fertilisers: Urea (containing 46 per cent N) and di-ammonium phosphate (DAP: 46 per cent P).

Fossilisation, moreover, isn't confined to tractors and nutrients. The building blocks for manufacturing most of the active ingredients in crop protection chemicals — insecticides, herbicides and fungicides — are benzene, toluene, xylene, ethylene and propylene, which are, in turn, derived from naphtha. The solvents, emulsifiers and surfactants, which dissolve, blend and stabilise these ingredients to create uniform sprayable formulations, are also largely petroleum-based.

The West Asia and Black Sea region conflicts have pushed up the prices of liquefied natural gas — the primary feedstock for urea and other N-containing fertilisers — delivered ex-ship to Northeast Asian markets from around $11 to $30 per million British thermal units, and that of Northwest European naphtha cargoes from $515-520 to $910-915 per tonne, over the past year.

Equally significant is sulphur. No production of DAP, SSP or any other fertiliser with P is possible without this byproduct of oil refining and natural gas processing.

Till 2021 or so, sulphur was imported into India at below $300 per tonne — even sub-$100 in 2019 and 2020. Landed prices rose to $400-500 per tonne during March-June 2022 after Russia's invasion of Ukraine. As Ukraine has stepped up drone and missile strikes on Russian refineries and gas complexes, and Iran targeted the energy infrastructure of its Persian Gulf neighbours, prices of the once-throwaway product have soared to $1,000-1,100 levels.

The risks from agriculture's heavy reliance on hydrocarbons wouldn't have been evident when India embarked on the Green Revolution in the mid-sixties. The iconic 'Mere Desh Ki Dharti' song from Manoj Kumar's 1967 blockbuster *Upkar*, indeed, had lines extolling traditional farming: *Bailon ke gale mein jab ghungroo jeevan ka raag sunate hain… Sunke rahat ki aawazen yun lage kahin shehnai baje* (when the bells hung around the bullock's neck sing the song of life; the sound of the *rahat* seems like that from the *shehnai*).

> The West Asia and Russia-Ukraine conflicts, together with El Niño, are testing the limits of both fossilisation of farming systems and the growing fuelisation of food

But it isn't fossilisation alone. What we are also witnessing in parallel is the fuelisation of food. The longer Brent crude prices stay at $100-plus per barrel, the more the incentive for diverting food crops towards biofuel production. This is reverse fossilisation, wherein agriculture becomes more a source than a consumer of fuel.

In India, we have ethanol produced from sugarcane, rice, maize and other starchy cereal grains being blended up to 20 per cent in petrol. Vegetable oils are, likewise, amenable for conversion into fatty acid methyl esters (FAME), which can be blended with regular petroleum diesel. Indonesia has, from July, made it mandatory for all diesel sold nationwide to contain 50 per cent palm oil-derived FAME.

If the ongoing wars are laying bare the vulnerabilities from fossilisation, the limits to the fuelisation of food could be tested by a prospective "very strong" El Niño. When the two supply shocks combine, it's agriculture that bears the brunt. The last time India experienced such a twin shock was in 1979-80, when the farm sector contracted by 12.8 per cent and its oil import bill doubled on the back of Iran's Islamic Revolution.

Indian agriculture is, of course, far more diversified now than in 1979-80, with a growing share of farm incomes coming from dairying, poultry, aquaculture and horticulture, as against normal field crops. Also, about 56 per cent of its gross cropped area is under irrigation, whereas that ratio was 29 per cent in 1979-80 and 17.1 per cent in 1950-51. The twin shocks from wars and El Niño are unlikely, then, to cause sharp farm output declines of the kind seen in the past.

That said, policymakers would do well to learn lessons from the still unfolding energy-cum-climate shocks. The implications for agriculture cannot be ignored when India is an importer of oil and gas, fertiliser raw materials (ammonia, rock phosphate, sulphur and muriate of potash) and technical-grade active ingredients and intermediates in crop protection chemicals.

There is scope firstly to reduce the consumption of urea and DAP, especially given their low nutrient use efficiency. Only a third of the 46 per cent N in urea is absorbed by crops; the rest escapes into the air as ammonia gas or leaches underground after conversion into nitrate. Even with the 46 per cent P applied through DAP, three-fourths of it simply gets "locked up" in the soil and is unavailable to the plant roots during the growing season.

The focus must be on gradually replacing urea and DAP with ammonium sulphate and complex fertilisers having more balanced nutrient composition with lower N volatilisation/leaching losses and P lock-up. Nutrient uptake and absorption can be further enhanced by biostimulants that expand the root surface areas and modify the metabolic pathways of plants. A similar approach — to trigger the internal defence and adaptation mechanisms of plants (think vaccines) — can bring down chemical pesticide application. Irrigation, too, should progressively move to solar-powered pumps and drip systems.

As for fuelisation, the Modi government has already banned the use of sugarcane juice and B-heavy molasses for making ethanol. As El Niño bites, the turn of grains may also not be far.

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On climate, a long road back to Paris

THE WRITING has been on the wall for at least three years: The Paris Pact's goal of keeping global temperature rise down to 1.5°C above industrial levels will be unattainable in the near term. The United Nations Environment Programme's (UNEP) "Emissions Gap Report 2026", released on Wednesday, underlines this. It sets out what it would take to return to the Paris threshold by the end of the century. Under a scenario designed to achieve the lowest possible overshoot, global warming would peak at 1.7-1.8°C around 2050 before returning to 1.5°C in the next half a decade. This pathway requires global GHG emissions to fall by 18 per cent by 2030, 45 per cent by 2035 and 69 per cent by 2040, relative to 2025 levels.

The scale of the challenge becomes clear when the projected reductions are measured against the climate action plans countries submitted under the Paris Agreement. The full implementation of these would reduce global emissions by only 5-8 per cent by 2030 and 11-15 per cent by 2035, relative to 2025 levels — well short of the reductions suggested by UNEP. The report mentions futuristic technologies like carbon storage. But the jury is out on their costs. The Emissions Report's renewable energy targets, too, are demanding, but solar and wind power growth in the past decade offer reasons for optimism. Installations have increased 2.5 times since the Paris Agreement. According to UNEP, they will need to surge by three times above 2025 levels in the next 10 years. Technological progress and falling costs can support such a transition. The more difficult task will be to shield the market from geopolitics, especially to ensure clean energy can be stored.

Climate-related damage cannot be ruled out, at least in the immediate future. The question, then, is how to compensate the most vulnerable. The shortcomings of the Loss and Damage Fund, formalised three years ago, were evident in its failure to aid Nepal after floods in August. When climate delegates meet at the next CoP in about two weeks, they will need to summon greater resolve than in the past to get the Paris Pact back on track.

For migrant children, access to school is not enough



KULDEEPSINGH RAJPUT

AS THE Kharif harvest comes to an end, seasonal migration begins across many parts of India. But this year, drought is adding a new dimension, with seasonal migrants increasingly becoming "climate migrants". Intense drought, drying fields and shrinking livelihood opportunities are likely to push more families towards sugarcane fields, brick kilns, construction sites and other worksites in search of livelihoods. The most overlooked victims of such migration are often children. Forced to migrate with their parents, they face interrupted schooling, learning loss, and the risk of prolonged educational exclusion. According to the last Census, India had 9.29 crore migrant children. At present, there is no reliable estimate of how many are out of school. The recent NITI Aayog report, 'School Education System in India: Temporal Analysis and Policy Roadmap for Quality Enhancement', highlights the educational exclusion of migrant children. Schooling disruptions, often lasting five to six months, widen learning gaps, weaken social ties, and undermine school belonging. These increase the risk of irregular attendance and dropout. Many children begin assisting their parents at work and slip into child labour.

The Sarva Shiksha Abhiyan, launched in 2001, marked a systematic effort to address barriers to school access among migrant children. Alongside expanding educational infrastructure in remote and underserved areas, it promoted context-specific alternative education models to ensure continuity, including seasonal hostels, residential schools, *basti shala*s, *sakhar shala*s (sugarcane schools), tent schools (construction-site schools), and transit education centres. Both the Right to Education Act (2009) and the National Education Policy (2020) have recognised the educational exclusion of migrant children and underscored the importance of inclusive education. Recently, NITI Aayog has recommended measures such as migration-responsive schooling, bridge learning centres, portable school records, inter-state transfer mechanisms, and tracking systems. Despite this, in practice, the primary focus has remained on access and enrolment.

Access and enrolment are only the first steps. The challenge lies in ensuring meaningful learning, remedial teaching, academic and mental-health support, emotional well-being, care, and equitable educational strategies. While providing education at worksites is a significant intervention, there is an equally urgent need to implement equitable and context-responsive alternative education models that address the distinct learning needs of migrant children. This is precisely what the NEP 2020 advocates.

> The challenge lies in ensuring meaningful learning, remedial teaching, academic and mental-health support, emotional well-being, care, and equitable educational strategies

The question is whether teachers have the pedagogical competencies and sociological perspectives to critically interrogate the sociocultural and structural inequalities reproduced in classrooms. Most teacher-education programmes are structured around the assumptions of sedentary formal schooling; migration-responsive education must be urgently integrated.

The true measure of an inclusive education system lies not merely in whether migrant children enter classrooms, but in whether they can learn, participate, and progress without their mobility determining their educational futures.

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